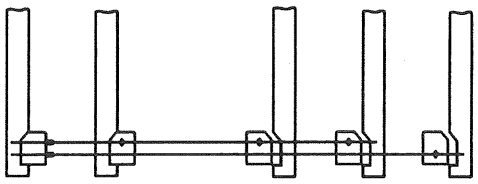
INTRODUCTION TO SIGNALLING

9

INTERLOCKING



A.J.R.Rowbotham

INTRODUCTION

The interlocking has to ensure that the railway behaves in a safe manner - that is:

- Conflicting and opposing routes cannot be set.
- Points are only moved when it is safe for them to do so.
- Signals only clear to a proceed aspect when all the safety conditions are met.

Not only must the interlocking ensure that the railway behaves in a safe manner, but it must ensure that failures do not provoke a less safe situation, e.g. it must be fail-safe. Fail-safe philosophy depends on the circumstances. In the aerospace industries a plane, for obvious reasons, must be kept flying under failure conditions. On the railways, however, trains are brought to a halt. The interlocking has to arrange that signals show a stop aspect, and that points cannot be moved under failure conditions. This philosophy does of course tend to annoy passengers trying to get home after a hard day at the office, but they do not realise that it is ultimately for their own benefit.

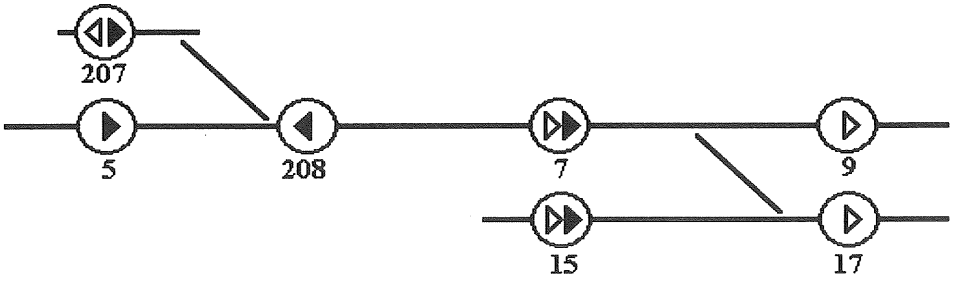
ROUTES AND ROUTE SETTING

Route Setting Process

In the days of mechanical lever frames each point and signal were separately controlled. To set up a route from one signal to the next, each point had to be moved to its correct position and finally a lever pulled to clear the signal. Interlockings can now work on a *route setting* principle, where one action sets the route, including all the point setting. The most common system in use today is known as the '*NX*' system. This is an acronym for eNtrance/eXit, as to set a route the signaller pushes a button associated with the signal at the start (or *entrance*) of the route, followed by a button associated with the signal at the end (or *exit*) of the route. This is the only action necessary, the interlocking taking care of setting all the points the correct way as required. The system will ignore combinations of button pairs where no route exists between them.

Each signal has an associated button, which is operated by a perspex rod. A lamp is incorporated in the base of the button, and when lit, the perspex rod acts like a fibre optic tube and appears to be illuminated. Engraved on the top of the rod are arrow heads indicating which direction of traffic is applicable to that button. A filled-in arrow head indicates that the button is at the entrance to a route, whilst an outline arrowhead indicates that it is an exit of a route.

The diagram below shows the principles involved.



The button at signal 5 is assumed to be an entrance only, for moves from left to right, as the signals behind 5 are automatics with no control by the signaller. The exit for the route from 5 is 7, so the latter button incorporates an outline arrowhead, reflecting this. However, 7 is also an entrance for a further route, and so also has a filled-in arrowhead as well, in the same direction. The button at signal 9 is an exit only, being assumed for this exercise that it is automatic with no signaller's control. The button at 208, a shunt signal, is an entrance for moves in the opposite direction, for backing a train stopped at 7 into the siding. The exit for this is at 207, which has an outline arrow pointing to the left. Signal 207 is also an entrance, in the opposite direction, for moves out of the siding up to the exit at 7. The two arrows on this button are in opposite directions - the exit arrow pointing into the siding, and the entrance arrow pointing out of the siding. Having pushed 7 as an entrance, the signaller has a choice of two exits, 9 or 17 to follow.

When a button is pushed as an entrance, the button light flashes as a reminder. It is possible to add a time-out feature that cancels the registered entrance and the flashing light if an exit button is not pushed within a certain time window.

The indications seen will be described in the relevant sections below.

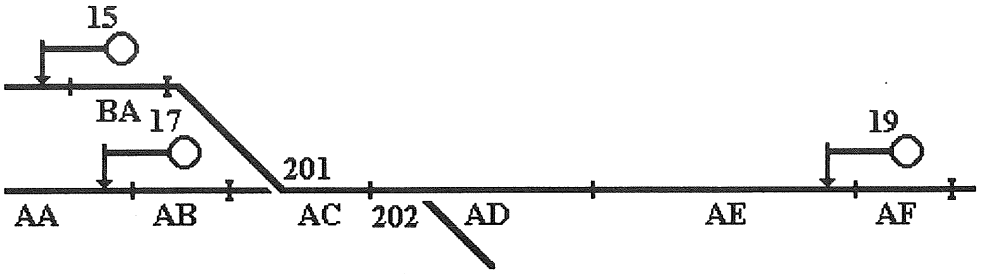
Integrated Electronic Control Centre (IECC)

The IECC is an electronic package encompassing all the non-vital functions required in a control centre. The signaller has a number of VDU displays, together with a keyboard and trackerball. The VDUs display a representation of the track layout, and alarm messages. To set routes the equivalent of pushing an entrance and exit button is performed, positioning the cursor with the trackerball and clicking. VDU indications are broadly equivalent to those described for hard panels below, except that coloured symbols are displayed rather than individual lamps. The whole of a coloured bar representing a piece of track can change colour to indicate different states, rather than having a number of lamps distributed along its length

The IECC also deals with other tasks such as Train Reporting, Platform Indicators, Public Address Announcements, Management Information etc. One particularly valuable task is that of *Automatic Route Setting.* If the IECC system has access to the computer holding the national timetable, then the 'system' can be organised to set routes directly from the timetable data without any action by the signaller. This relieves signallers of their mundane tasks, and lets them concentrate on sorting out problems. The 'system' ultimately has to convert the basic information from the timetable into the setting of individual routes. As an example the timetable might say that a train leaves Euston station at 10.00 and arrives at Watford Junction at 10.15. There are fast and slow lines all the way, with crossover facilities at both Willesden and Harrow & Wealdstone. The system has to decide whether to use the fast or slow lines for each section, and a preferred option can be defined. Then each section has to be broken down to the individual signal routes. However, the whole path from Euston to Watford cannot all be set at the same time. This would stop any other trains using the line and bring a large area to a halt. The 'system' has therefore to set the routes over a period of time in advance of the train's position, in sufficient time so that the driver does not see any restrictive aspects, but not far enough ahead to unduly block the line for other movements.

Route Availability

When a valid route request is received, the interlocking first checks the *availability* of each set of points involved in the route and overlap. Points are deemed to be available if they are already lying the correct way, or are NOT locked the other way. Only if <u>all</u> the points are found to be available will the route setting process continue and points be driven to the required position. This full availability check stops any points being driven unnecessarily if part of the route is not free to be used. Considering the layout



If a route from 17 to 19 signal is being set, the entrance signal (17) button light will start flashing when the button is pushed. The route is defined when the exit button (19) is pushed, and the interlocking would first check that there is a physically possible route between entrance 17 and exit 19. There is no physical route between 15 as an entrance and 17 as an

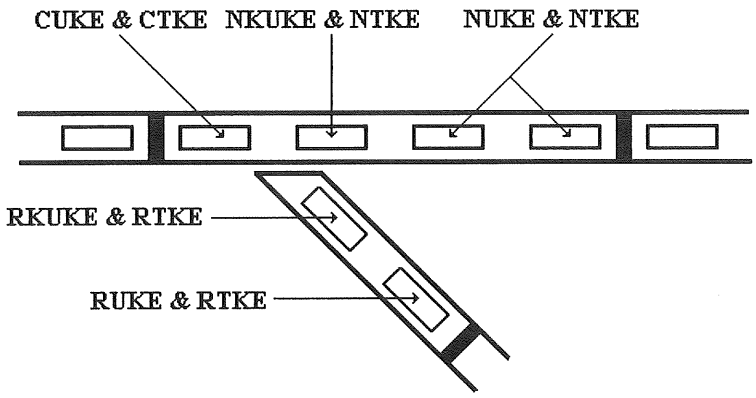
exit, so if this combination was pushed the system would ignore the request and cancel the registered entrance. Next the availability of both 201 to be reverse and 202 to be normal would be checked. If both are available the points will be driven to the correct lie if they are not already in that position.

Route Locking

While the points are being set, a locking function is set up for each track circuit along the route and overlap. In the example, such functions would be set up for track circuits AB, AC, AD, AE and AF. This locking function is provided for each track circuit even if there is actually nothing to lock, such as AB track. The function for AC will lock 201 points and that for AD will lock 202 points. These points are then *route locked* so they cannot be moved as long as the route is set. Thus 201 will be locked reverse and 202 will be locked normal.

The locking function being applied to each track circuit also drives white *route lights* along the line of the route on the panel. The appearance of the white lights reminds the signaller of which routes are set, and their illumination also proves that the route is properly locked. When the route is locked and the route lights are fully illuminated the light in the entrance button becomes steady.

As each track circuit becomes occupied by the passage of the train, the white route lights are extinguished, to be replaced by red *track lights*. Red lights along the line of track thus indicate the position of trains. To ensure that route and track lights are only lit along the line of route, the functions have to be split into three parts.



The *common route lights* (CUKE) are required whichever way the route is set, and are illuminated with either the *normal route lights* (NUKE & NKUKE) or the *reverse route lights* (RUKE & RKUKE) as appropriate. The first of each of the route lights past the divergence also prove the point detection (NKUKE and RKUKE), so that if the points are having to be driven to their required position, the appropriate light will be illuminated later than the rest of the UKEs. Similarly the track lights are split into three, the common (CTKE), the normal (NTKE) and the reverse (RTKE). Normally, track lights when illuminated replace the route lights appropriate to the route that is set. However, if a track becomes 'occupied' without a route being set first (probably because of a track failure), then the common, normal and reverse track lights are all illuminated together to draw the signaller's attention to the problem.

Train Operated Route Release

In order to release the route locking following the passage of the train, it is necessary to first cancel the route. This is normally done by the passage of the train past the signal. The technique, known as *Train Operated Route Release (TORR)*, relies on a sequence of track circuit operations, rather than just one becoming occupied. Using a single track circuit would not be very safe as a track circuit failure could release the locking in front of a train. The TORR requirements have to be looked at for each signal individually as there are many variations in the conditions. A typical basic TORR release would be the sequence of AA track occupied with the signal showing a proceed aspect, followed by AA clear with AB occupied and the signal replaced, i.e. the whole train has passed the signal. More complex TORR releases might well involve a sequence of three track circuits. When the TORR cancellation of the route is achieved the light in the entrance button is extinguished.

Release of Route Locking

Once the route has been cancelled the locking is removed track by track behind the train in sequence as it passes along the route. Taking track AD as an example, its locking can only be removed if the locking has already been removed from the previous track, AC, and if AD track itself is clear. Consequently 201 is unlocked and may possibly be used for another route when the whole train is clear of the points, i.e. AC track is clear. Similarly, 202 is released when AD track circuit is cleared behind the train. This technique is known as *sectional release route locking*. The sequential releasing method is the reason for a locking function being set up for every track circuit, irrespective of whether there is actually anything to lock.

As the lock is removed from each track circuit, the white route lights will also be extinguished, though in practice the white route lights will have been overridden by the red track lights. As each track circuit is cleared the red track lights will disappear and there might be a short flash of white lights before the panel is left blank.

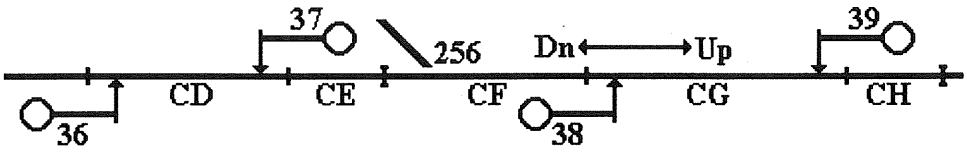
In this particular layout 201 cannot be used for any other route when AC becomes clear with the train still on AD track. However both points could be used for a route from 15 signal through 202 reverse when AD becomes clear. It follows that money has been wasted in installing two track circuits over the two points. One track circuit, covering both points, would have given all the flexibility possible in this particular layout. It is this sort of consideration that determines how many track circuits are necessary in point areas.

Conflicting Routes

If the route is set from 17 signal to 19, then it must not be possible to set any route from 15, as any route from 15 is a *conflicting* route. The fact that the route is set from 17 to 19 means that 201 is route locked reverse. Hence, if any route from 15 is attempted the availability check (for 201 to be available normal) will fail and no route will set. The prevention of conflicting routes is thus done automatically in the point availability and route locking functions. There is no need to formally lock 15 and 17 signals against each other.

Opposing routes

Routes which conflict with each other, but do not require points to be set in different ways for the two routes, usually involve traffic moving in opposite directions, and are known as *opposing* routes. As the point availability checks have no effect in this situation, something formal has to be done instead.



If a route is set from 37 to 39, then 38 to 36 is an opposing route (and vice versa), and it must not be possible to set it. Points 256 are required normal for both routes so cannot be used to prevent the opposing route being set. The locking function set up on each track circuit during the route setting process is in fact directional. For instance setting a route from 37 to 39 sets an up-direction locking function on track circuits CE, CF, CG and CH. Conversely, setting a route from 38 to 36 sets a down-direction locking function on CF, CE, CD etc. Having set a route from 37 to 39, any route from 38 must be locked out until the route from 37 has been cancelled and the train completely passed 38 signal. Only then could a route be set from 38 in the down direction. This release condition will only occur when the up direction locking function set on CF is removed in the normal sectional release of the route. Thus when a route from 38 is requested it can only set if at least there is no up-direction locking on CF track. Similarly, routes from 37 signal can only set if there is at least no down-direction locking on CE track.

Manual Cancellation and Approach locking

A route that is set may be cancelled at any time by pulling the entrance button. The signal will immediately then be replaced to its most restrictive aspect. However, it might well be dangerous to actually release the route locking immediately. If a train was approaching the signal at speed and was perhaps a few hundred meters away from the signal when the route was cancelled, it certainly would not have sufficient braking distance to stop at the signal. If the route locking had been released the train would pass the (red) signal and enter the unlocked route before it could stop. There would be nothing to stop the points being moved, except dead locking (see below), and the train could easily be derailed or points damaged.

This problem is solved by keeping the route locked for a period of time, typically two or three minutes after the button is pulled. During this time either the train will have stopped before reaching the signal, or will have passed the signal and entered the route. Once the train has entered the route, the locking will be held in front of the train as described above, even when the time period elapses.

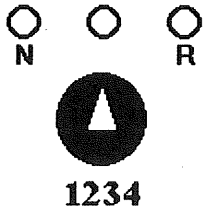
If there was no train approaching the signal when the button was pulled, it would be rather restrictive to have to wait for the timing period, so it is possible for the system to 'look back' to see if a train is approaching or not. If there is no train, the route can cancel immediately, if there is a train approaching then the timer must be invoked. However not all interlockings have this feature, and so have to apply the timer every time. These techniques are know collectively today as *approach locking*.

POINTS

Individual operation

The position of the points is normally controlled by the route setting process. However, means are provided for the signaller to individually control points if required. This individual control might be used if the route setting system has failed, or if the points are undergoing maintenance or testing.

Each point has a numbered three position switch, or '*point key*', accompanied by three indications.



The key is usually in the centre position, and in this position the route setting process can function unhindered. If no routes are set, moving the key to the normal position, and then returning it to the centre position, will drive the points normal (if they are not already lying normal). Similarly, the points may be driven reverse by turning the key to the reverse position and returning it to the centre position. If a route is set, locking the points, then operation of the point key will have no effect.

If the switch is left in the normal or reverse position, rather than returning it to the centre position, then the points will be locked in that position. If the point key is left in the normal position, locking the points normal, a normal route may still be set through the points without problems, but it would not be possible to set a route requiring the points reverse. Returning the key to the centre position releases the key locking.

The three indications are coloured white, and the outside ones show the position of the points by illuminating either the normal (N) or reverse (R) indication. If no detection is being received back from the relevant point machine, then the centre, *out of correspondence* light flashes, thus rapidly drawing attention to the problem. Every time the points are driven there will be no detection for two or three seconds while the points are moving, so the indication will flash during this short period. Some customers like to suppress this flashing during normal movements by delaying the indication by seven seconds. If this is done, the light will only flash when there is a real failure.

Dead Locking

It is important that points must not move under a train, so if the track circuit over the points is occupied, then the points must be locked, and so cannot be moved by route setting or by the key. This feature is known as *dead locking*. Normally for a train to be moving over a set of points a route must be set, so in fact the points will be locked twice while a train is traversing them - once by the route locking and once by the dead locking.

ASPECT CONTROL (MAIN SIGNALS)

The last part of the route setting process is the clearance of the entrance signal to a proceed aspect if the conditions are correct. What that proceed aspect of a main signal is, will depend on the number of aspects on the signal, and the aspect being shown by the signal in advance.

Main signals are dealt with first, followed by the requirements for shunt signals.

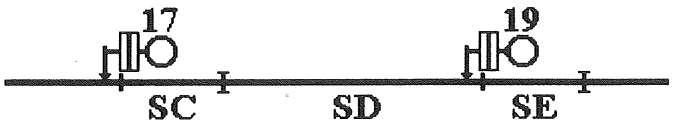
Lamp Proving

If the main signal lamp required to be illuminated were to fail, then the signal would be dark with no aspect showing. This could lead to a dangerous situation, especially at night when the physical signal is not easily seen. It is a requirement that each signal can only show a proceed aspect if the signal in advance is in fact properly showing an aspect. This requirement is generally known as *lamp proving* and has to be fail-safe in operation

To improve the reliability, special lamps are used, each with two filaments. If the first filament fails, then a non-vital relay in the signal head switches in the second filament. At the same time, an alarm is sent back to the interlocking, warning of the failure and the need to replace the lamp. This *filament failure* system has no effect on the signal in rear.

Automatic Signals

If there is only plain track between one signal and the next (including the overlap at the exit), then the signal can be arranged to work automatically without any action required by the signaller. The symbols for automatic signals on panels are designated by adding a rectangle representing the actual number plate of an automatic signal to the post symbol.



On such plain track there is no need for a number of track circuits, one track circuit plus the overlap track would suffice. For 17 to show a proceed aspect, three track circuits, SC, SD and SE must all be proved to be clear. Signal 19 must also be proved to be lit. Thus the condition for 17 to show a proceed aspect is:

```
SC clear AND SD clear AND SE clear AND 19 lit
```

If any of the four items in this list is not valid, then the signal will immediately revert to a red aspect. Directly the front of the train enters SC track the signal will be replaced to red, and it will not clear again until the rear of the train has cleared SE track.

In long sections of line with automatic signals it is possible to simplify the layout even more, with just one track circuit per signal.



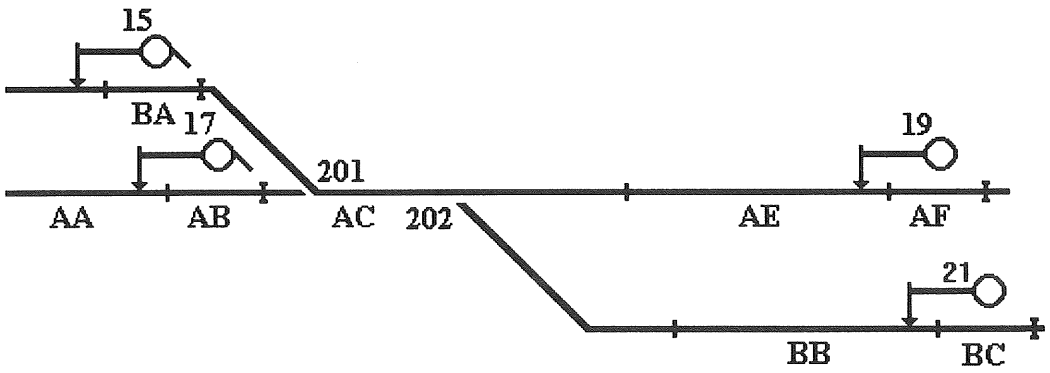
Now the condition for 13 to show a proceed aspect is solely:

```
SH clear AND 15 lit
```

Automatic signals are not indicated on the panel, so the signal symbol is just paint, but the signaller may have a control to replace the signal to danger in an emergency (*emergency replacement*).

Controlled signals

If there are any points in the route or overlap, then the signal has to be controlled status. A controlled signal is indicated on the panel,. with, at the time of writing, only red and green indications given, the latter representing any proceed aspect. The absence of an indication would indicate that the signal lamp (or the indication lamp) has failed.



Considering firstly the route from 17 to 19, the signal must only clear when all the appropriate track circuits are clear and the exit signal lit, as before. In addition any relevant points must be detected the correct way. Finally it must be proved that the route is set and locked. The condition for 17 to clear for a route to 19 is:

```
Route set AND AB clear AND AC clear AND 201 reverse
AND 202 normal AND AE clear AND AF clear AND 19 lit
```

It is convenient to list the various items in order along the route, although the actual order in a logical statement like this is not relevant.

The logic has a *one train* feature that requires the route to be set for every train following the TORR for the previous train. However, it is possible for an additional control to be provided on the panel, an *auto button*, that overcomes this feature to allow the signal to work automatically once the route is set and locked.

Approach Control

Considering now the route from 17 to 21, similar arguments apply - all the tracks must be proved clear, points detected the required way and the exit signal lit. However, an additional feature, *approach control*, is required.

This route involves a move through 202 reverse, and such a divergence would usually be a slower speed move then the 'straight' move from 17 to 19. The object of approach control is to force the driver of a train to reduce speed to a safe level for this divergence. When the route sets, the signal does not actually clear even though the route is properly set and locked. The train consequently approaches 17 at red and will be slowed to stop appropriately by the driver. However, when AA track is occupied the route arm and signal will clear. Even if the driver then accelerates at the maximum possible rate, the speed of the train will not be excessive over the reverse points. There is nothing on the signal as the driver approaches to indicate that the route is set, and the driver dares not anticipate the clearance as he approaches as the route might not in fact be set. The length of AA has to be calculated to achieve the required reduction in speed. Alternatively, the requirement might be for the track to be occupied for a certain period of time before the signal clears.

The condition for 17 to clear to 21 is:

```
Route set AND AB clear AND AC clear AND 201 reverse AND 202
reverse AND BB clear AND BC clear AND 21 lit AND AA occupied
```

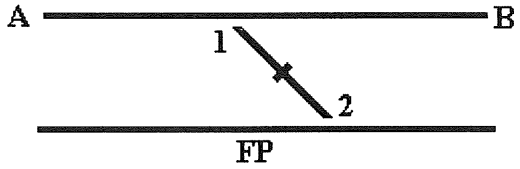
There are consequently two sets of condition for 17 to clear, i.e. either the set for 17 to 19, OR the set for 17 to 21. Some items are in fact common to the two routes, so the requirements for 17 to clear can be succinctly written as:

```
Route set AND AB clear AND AC clear AND 201 reverse
AND [(202 normal AND AE clear AND AF clear AND 19 lit)
 OR (202 reverse AND BB clear AND BC clear AND 21 lit AND
                                          AA occupied)]
```

Notice the nested brackets around the expressions to clarify the precise logic required.

FLANK PROTECTION

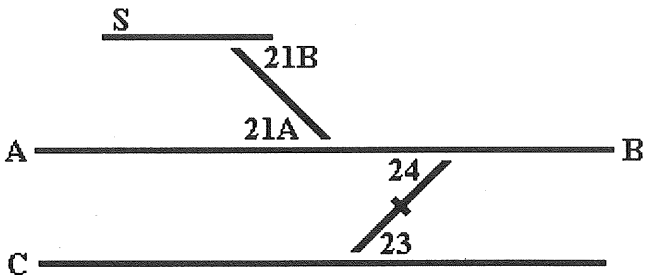
It is sometimes necessary to incorporate in the controls an item that is not on the direct line of route. This technique is known as *flank protection* as it is used to protect the flank, or side, of the train. Considering a crossover between two parallel lines:



A train travelling from A to B on one line through 1 points normal must be protected from a movement on the other line with 2 points reverse. For many years the two ends of the crossover were worked as one, so either both ends were normal or both ends were reverse. In this way flank protection was automatically provided. However, this traditional arrangement leads to unreliability. Both ends of the crossover have detection and the two are logically ANDed together to indicate both ends normal or both ends reverse. This however means that if the detection fails on one end, both roads would be blocked.

To improve the situation the two ends may be numbered separately, with individual detection. When a route is set from A to B, points 1 must be conventionally set, locked and detected normal. The same route also requires that points 2 are set and locked, but not detected, normal. Any failure of detection on 2 points does not then directly affect the route from A to B, although flank protection is provided. However, this situation does lead to the worry that if the detection of 2 points does fail, the points might be lying reverse, so as an added complication, FP track circuit has to be proved clear in the route from A to B if 2 points are not detected normal.

Another classic case of flank protection is shown below:



Movements from A to B would be protected from unauthorised movements out of the siding S by the double ended points 21 required to be set, locked and detected normal. However, moves from C to B (or B to C) would need similar protection, so this route would also require 21 points, not on the line of route, to be set locked and detected normal.

OVERLAPS

All the examples so far have shown an overlap consisting of just one track circuit. However overlaps may well involve more than one track circuit and also points.

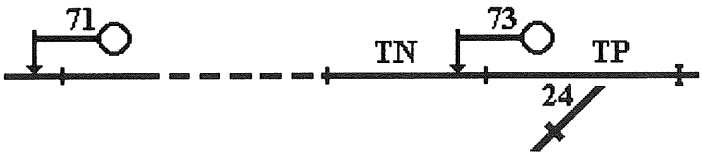
Considering more than one track circuit in the overlap:



The aspect controls for 71 signal must include both tracks clear.

```
....AND AR clear AND AS clear
```

If there are trailing points in the overlap they are treated just as if they are in the route between the signals, i.e. the points are set, locked and detected.



In this case the aspect controls must include the overlap track and point conditions

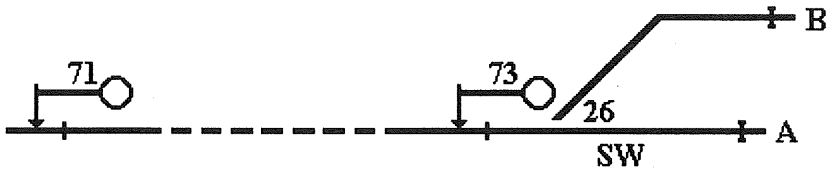
```
....AND TP clear AND 24 normal
```

The usual route locking is extended into the overlap to lock 24 points. One way of releasing this overlap locking would be for the 'forward' route from 73 to be set, the train to pass over the points and release the overlap locking when the rear of the train has cleared TP track in the usual way. (Note that when the forward route is set, the points will be locked twice - overlap locked normal by 71 signal and route locked normal by 73.) However a second release is necessary to stop the line jamming to a halt.

Suppose 73 signal was on the end of a platform. A train could pull into the station and wait for perhaps a quarter of an hour. The points 24 would then be locked normal for that period of time, stopping any possible moves over 24 points reverse. An alternative release is given if the route has been cancelled and TN track circuit has been occupied for a period of time, perhaps 90 or 120 seconds. An occupancy of this length is deemed to indicate that the train is stationary, and so it is safe for the overlap locking to be released.

Swinging Overlap

Consider an overlap containing a facing point:

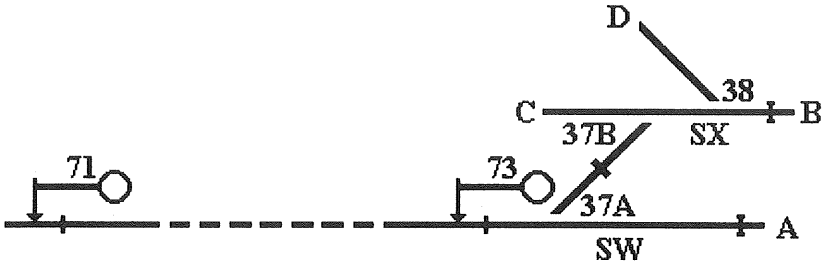


There is now the possibility of alternative overlaps. If 26 points are normal the overlap will be to A, and if the points are reverse the overlap will be to B. Normally the overlap is set the way the points happen to be lying when the route is set. The aspect control does prove that the points are detected one way or the other.

```
AND SW clear AND 26(normal OR reverse)
```

The forward route might be from 73 in the A direction requiring the points normal, or from 73 in the B direction requiring them reverse, so the points must not be locked. The action of setting the forward route will set (and route lock) the points as appropriate. This situation is known as a *swinging overlap* as the points may be swung from side to side in the overlap. There must be a time delay built into the aspect controls for the 26 normal or reverse condition so that 71 does not go back to red whilst the points are swinging. Points 26 can also be swung in the overlap with the individual points key if required.

Swinging overlaps can become quite complicated as in the example below.



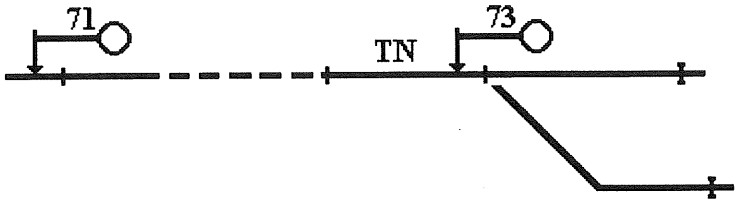
If the overlap is with 37 reverse, then more track circuits and points become involved. If 37 is lying reverse in the overlap then both SW and SX must be proved clear, and 38 will need to be set and locked normal. If the overlap was set with 37 normal and 38 reverse then swinging 37 reverse on the key would also drive 38 normal. If 38 were in fact locked reverse for any reason, then 37 points would not be able to swing reverse. The aspect controls are written in a slightly backwards fashion, using OR rather than WITH, that suits they way the requirements are actually achieved.

```
....SW clear AND 37(normal OR reverse)
    AND [(SX clear AND 38 normal) OR 37 normal]
```

If 37 were to be lying reverse, the action of setting a route (or overlap) from C to B or from D to B, will swing 37 points normal.

Time of Operation Locking

If the facing points are extremely close to the insulated joint at the signal, problems can occur.



If a train was running past 73 signal (at red) and the points were swung for some reason an instant before the front wheels crossed the joint, then the points could still be moving when the front wheels actually reached the toe of the points. (Dead locking by the occupation of the points track circuit allows the points to continue moving if they started before the dead locking track was occupied.) The technique to prevent this is known as *time of operation* locking.

The points are additionally locked by TN being occupied, until either the track is clear again, or the train is proved to be stationary by the track being occupied for a specific time. For the points to be free to move the logical condition is thus

```
....(TN clear OR TN occupied for 90 secs)....
```

ASPECT CONTROL (SHUNT SIGNALS)

The aspect control of shunt signals is not necessarily so rigorous as that for main signals. The detection of all points is proved in the aspect, but not necessarily every track circuit up to the exit signal.

The extent of track proving depends on the purpose of the shunt signal. Some might be for the sole purpose of moving an engine onto the front of a train. In this case the last track circuit in the route, where the train is standing, could not be proved clear.

Very occasionally it is a requirement to lamp prove a shunt signal, and sometimes it is a requirement for shunt routes to have overlaps. Each signal has to be considered as a separate case.

On the panel, the stop aspect of a shunt signal is indicated by a red light, and the proceed aspect by a white light.

GROUND FRAMES

There are many occasions where points, perhaps an emergency crossover or a rarely used siding connection, are not controlled directly from the interlocking, a cheaper method, the *Ground Frame* being used. The points, and sometimes shunt signals, are controlled locally from a small leverframe, or more usually today, a switch panel. To ensure safety, the local control can only be put into action when a *release* is received from the signaller. Until this release is received the ground frame controls are all locked, allowing normal running to take place.

The release can only be given if the protecting signal(s) are replaced, and they are *free of approach locking*, i.e. with no trains approaching them. If the Ground Frame is in an area of automatic signalling, then the protecting signals are termed *semi-automatic,* that is they normally work automatically, but can be replaced to danger by the signaller when it is required to give the ground frame release.

Once the release is received at the Ground Frame the points and signals may be operated as required with any necessary interlocking being provided at the frame. When the movements have finished, all items worked from the Ground Frame must be proved to be back in their normal position before the signaller can cancel the release and revert back to normal running.

CONTROL TABLES

The requirements for interlockings are recorded in *Control Tables*, a series of standard tabular format sheets individually filled in for each signal route and point.

Entries in the tables may be elaborated by the use of standard *'$ Notes'*. For instance:

- $7 Temporary Remote Control.
- $37 Points Set and Detected.

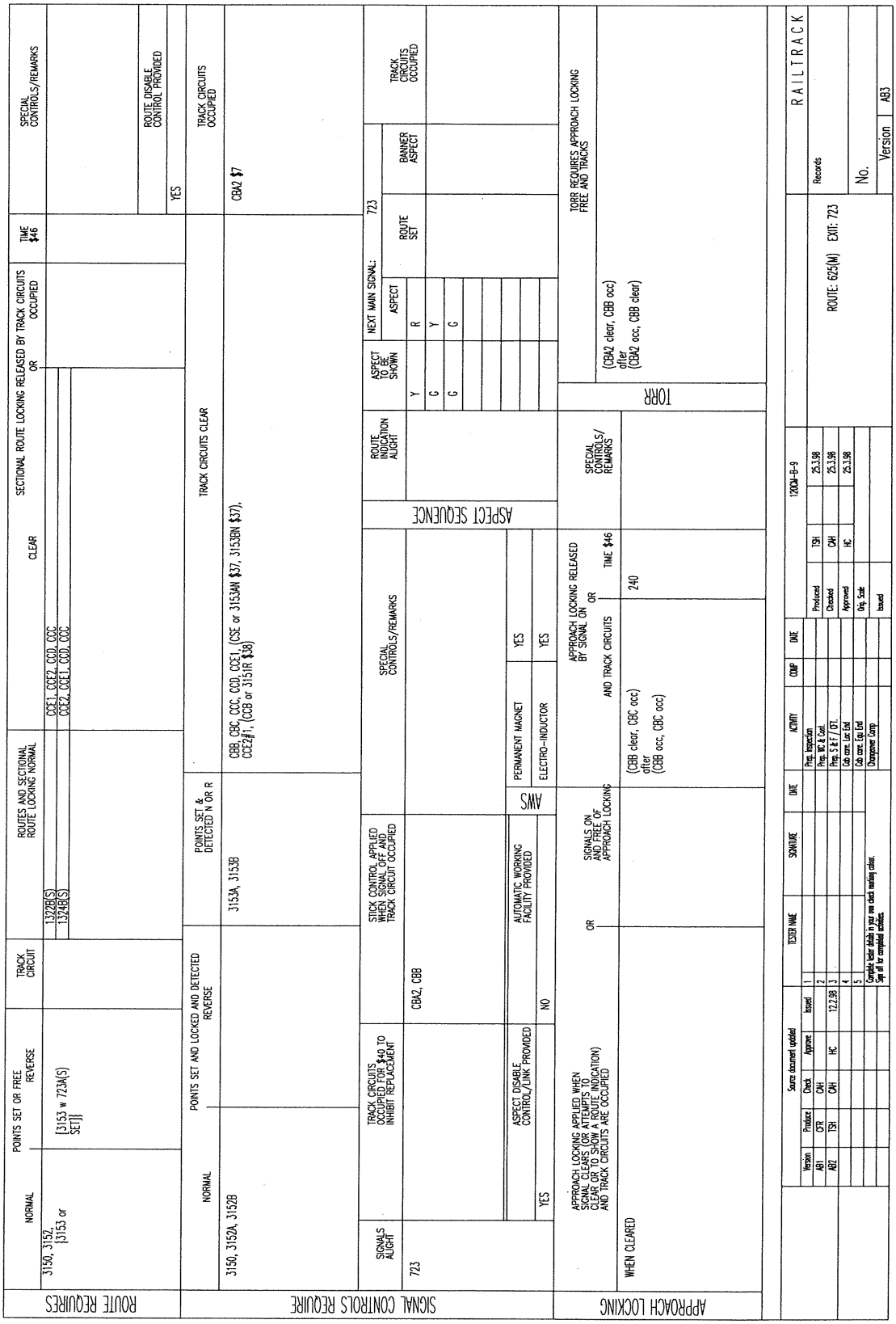
Figure 1 shows a typical sheet for one route of a signal. Each route from every controlled signal has an individual sheet defining the requirements:

- The first row shows the requirements for the setting of the route, with areas for point availability and opposing route locking.
- The second (double height) row across the sheet shows the requirements for the aspect controls, with areas for point detection and track circuits on the upper part. On the lower part are the areas for lamp proving, signal replacement conditions, AWS and aspect sequence requirements.
- The bottom row gives the approach locking and TORR requirements. The entry 'WHEN CLEARED' indicates that if this route is manually cancelled there is no 'look back' to see if a train is approaching, so the route will remain locked for a period of time, shown in the time column.

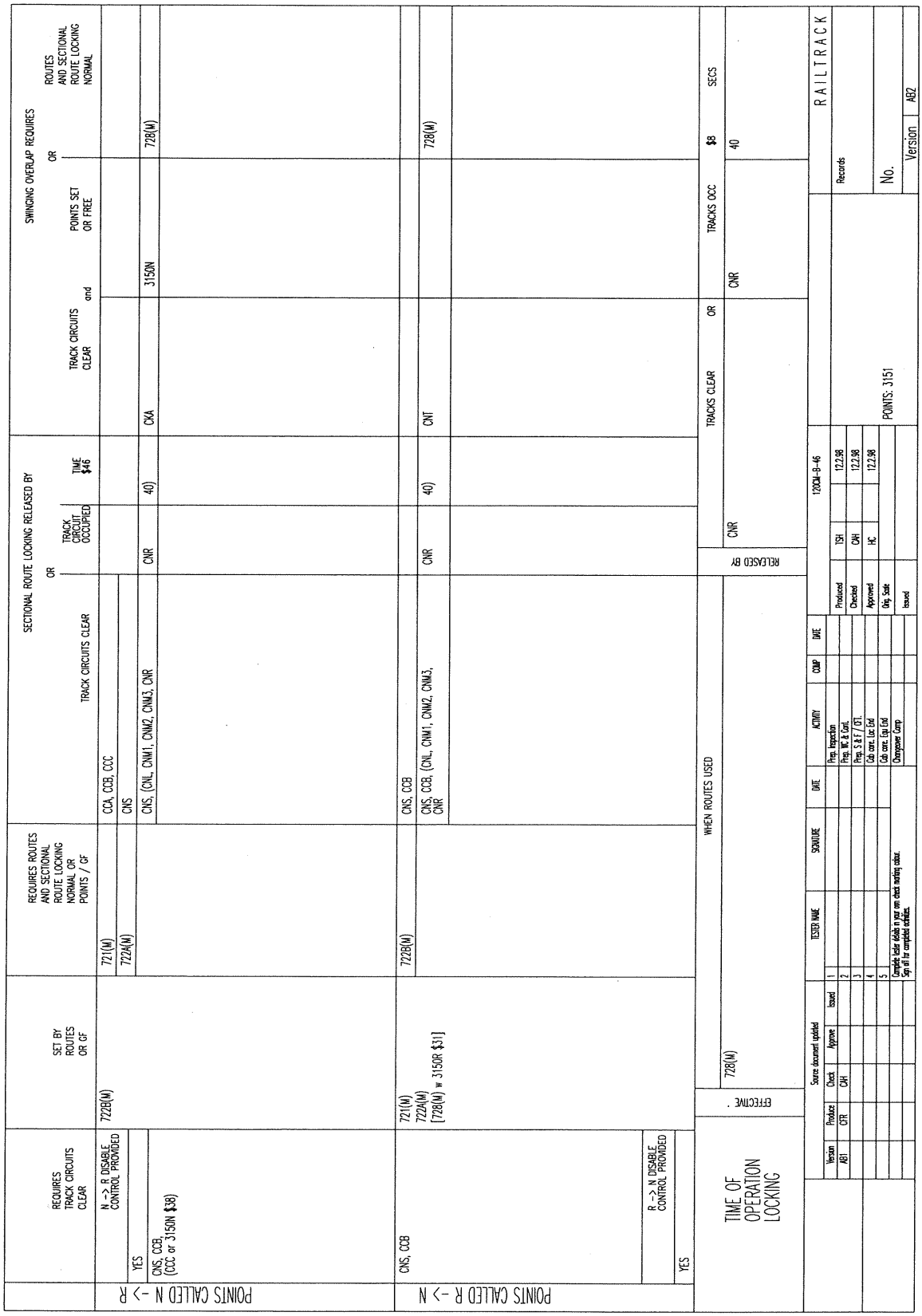
Figure 2 shows a typical point sheet dealing with both the movement from Normal to Reverse in the top part, and from Reverse to Normal in the next lower part. In complicated interlocking areas with a lot of entries, these two parts have to be shown on separate sheets:

- The first main column shows the dead locking controls.
- The second column shows those routes setting and locking the points reverse in the top part and normal in the lower part.
- The third column shows which routes (locking the points the opposite direction) have to be normal to allow the points to move to the reverse and normal positions.
- The next main column shows the releases required to overcome the locking in the previous column.
- The right-hand most column shows the swinging overlap requirements.
- Across the bottom of the sheet are the time of operation locking controls.

Other standard format sheets are available for items such as automatic and repeater signals (several on a page as their controls are relatively simple), special AWS controls, Ground Frames, different types of level crossings etc.



Typical Signal Control Table



Typical Point Control Table